

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap as the operating interval (BCa, $B = 10,000$, re-run on the full 60-experiment grid). The study was governed by a pre-specified protocol with registered, dated amendments, publicly archived with per-file checksums (see the Data availability statement). The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified empirical direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60

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(75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met ($23/60 = 38.3\%$), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit ($+0.0159$ vs. $+0.0083$). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and a discrimination-aware gate (C5) shows 7/14 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

Why this matters for ECG. PTB-XL [3], Chapman-Shaoxing [4], and CPSC differ in acquisition protocols, lead sets, and label vocabularies. A model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [23]. Clinicians and regulators need a decision rule: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

48 **Temperature scaling and pre-registration.** Temperature scaling
 49 (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted
 50 under a convex loss, it minimizes NLL on the calibration split, so its cali-
 51 bration benefit there is empirically small; on a shifted target the source-fit
 52 temperature is mismatched to the target logits and a non-trivial benefit can
 53 remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant
 54 under confidence-shifting monotone maps, and TS can *increase* ECE where
 55 the model is under-confident (nine negative cells in our data). It motivates
 56 the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned
 57 decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing,
 58 protocol v2.1-A1 [24] fixes the primary endpoint, test, family structure, and
 59 failure branches before unblinding; revision A1 re-locates the endpoint from
 60 **decay** to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with
 61 before/after protocol hashes [25, 26]. Multi-seed validation (5 seeds $\times$ 6
 62 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed
 63 results, including nine counter-examples, are reported transparently.

64 **Contributions.** Five independent contributions:

65 **C1 (Primary OOD benefit):** On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$,
 66 TS yields a statistically significant positive OOD benefit in 51/60 seed
 67 experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30,
 68 ResNet1D 24/30), with nine counter-examples transparently disclosed
 69 and mechanistically explained. InceptionTime: three of six directions
 70 achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two
 71 4/5, two 3/5.

72 **C2 (Boundary condition, pre-registered branch triggered):** Across
73 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI)
74 have 95% CIs containing zero and 34/60 have CI lower bound > 0
75 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD
76 mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI
77 containing zero) is *not* met, triggering the “ID nonzero boundary”
78 branch (protocol A1.4.2): the premise that TS has no ID recalibrat-
79 able miscalibration is re-examined, the decay contrast restored as
80 exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the
81 ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID
82 point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$
83 (counter-example 6, -0.0152 ; one marginal case, -0.0128).

84 **C3 (Methodological, Shapley decomposition validation):** A 27-
85 cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence,
86 $n = 20,000$) recovers ground truth in the identifiable regime (27/27
87 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a
88 threshold, and reports bootstrap CIs on absolute Shapley values.

89 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6
90 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior
91 adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included
92 (InceptionTime 15/30); after exclusion the EM OOD finding attenu-
93 ates. The mechanism is prior-likelihood mismatch: EM fits a target
94 prior that amplifies shift-induced miscalibration when the source con-
95 fusion matrix is ill-conditioned.

96 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer
 97 gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from
 98 clinical viability (accuracy $\geq$ source-domain baseline). Across 14
 99 pair $\times$ architecture strata, 7/14 pass both layers; 7/14 fail the discrimi-
 100 nation layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy
 101 below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$,
 102 $\text{acc} = 0.657 < 0.713$). The gate reframes the 9/60 counter-examples:
 103 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures
 104 (C5 rejects despite C1 passing).

105 Prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but
 106 not an actionable *when does TS pay off* answer with multi-seed robustness
 107 evidence. The novelty is *not* multi-seed validation on ECG (a methodologi-
 108 cal choice) but the **first systematic characterization of the boundary**
 109 **conditions under which TS yields a positive OOD calibration ben-**
 110 **efit in cross-corpus ECG transfer:** the pre-registered OOD quantifica-
 111 tion, the ID boundary condition, the method-selection boundary (TS robust;
 112 EM prior and Matrix scaling weakest), the predictability failure, and the
 113 discrimination-aware gate. That characterization is the actionable contribu-
 114 tion; multi-seed validation is the evidence standard.

115 2. Related Work

116 2.1. Calibration Estimation and Recalibration

117 The calibration of modern neural networks has been studied extensively
 118 since the seminal work of Guo et al. [2], who showed that deep networks
 119 are often overconfident and introduced temperature scaling (TS) as a simple

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

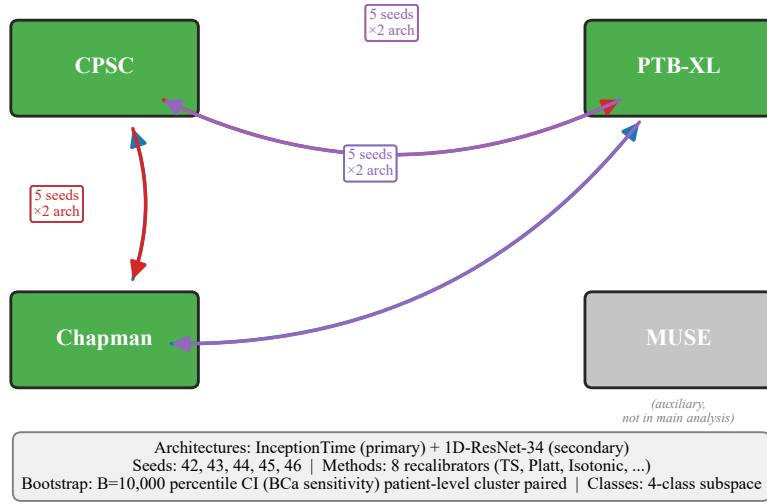


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

120 post-hoc recalibration method. Nixon et al. [6] proposed the static calibration
 121 error, and Kumar et al. [16] introduced verified uncertainty calibration
 122 with distribution-free guarantees. Kull et al. [7] generalized TS to Dirichlet
 123 calibration for multi-class settings. In the clinical risk-model context, Van
 124 Calster et al. [14, 15] established a calibration hierarchy from utopia to em-
 125 pirical data and argued that calibration is the “Achilles heel” of predictive
 126 analytics. Strodt Hoff et al. [17] benchmarked deep learning for ECG analy-
 127 sis on PTB-XL and noted calibration degradation under subpopulation shift,
 128 motivating our cross-corpus design.

129 *2.2. Prior-Shift Estimation and Transfer Calibration*

130 Lipton et al. [9] introduced black-box shift estimation (BBSE) for detect-
 131 ing and correcting label shift using confusion-matrix inverses. Alexandari
 132 et al. [8] showed that maximum-likelihood calibration with bias correction
 133 is hard-to-beat at label shift adaptation, providing the theoretical founda-
 134 tion for our EM prior-shift method. For transfer calibration upper bounds,
 135 TransCal [20] estimates transferability for auto-selecting target-domain un-
 136 supervised optimal T ; PseudoCal [21] uses pseudo-labels for calibrated trans-
 137 fer; and LaSCal [22] provides label-aware shift calibration with distribution-
 138 free guarantees. Our S1 paradigm differs fundamentally: we fit on the
 139 *source* calibration split only (no target labels), with S1’ prior-shift adaptation
 140 (EM/BBSE) as the only target-domain signal, a strictly weaker assumption
 141 than TransCal/PseudoCal, which access target logits or labels.

142 2.3. Cross-Domain ECG Calibration

143 Barandas et al. [10] evaluated uncertainty quantification methods in multi-
144 label ECG classification, finding that ensemble-based methods outperform
145 single-model approaches under subpopulation shift. Vranken et al. [11] stud-
146 ied uncertainty estimation for 12-lead ECG deep learning and reported signif-
147 icant calibration drift across acquisition devices. Haekal et al. [18] performed
148 cross-dataset validation for atrial fibrillation detection with calibration-aware
149 probability analysis, and Patel & Beedala [19] documented calibration drift
150 under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of
151 TS has been studied in general vision [1]; we contribute the multi-seed ECG
152 quantification and the boundary-condition analysis that identifies when TS
153 helps (ID) versus when it hurts (OOD).

154 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

155 Recent work has introduced ECG foundation models trained on millions
156 of recordings. ECGFounder [27, 35] (10 million ECGs, Harvard–Emory)
157 reaches expert-level performance on 80 diagnoses and transfers cross-domain
158 via fine-tuning. ECG-FM [28] is an open-weight transformer pretrained
159 on 1.4 million ECG segments with hybrid contrastive and generative self-
160 supervision, showing strong cross-dataset generalizability and label efficiency.
161 HeartLang [29] treats heartbeats as words and rhythms as sentences via
162 a QRS-Tokenizer and masked sentence pre-training. Tan et al. [30] found
163 that fine-tuning benefits diminish as the downstream dataset grows and that
164 CNNs transfer more effectively than RNNs in multi-label ECG classification.
165 For OOD adaptation, ADAPTOOD [31] quantifies distribution-shift sever-
166 ity via uncertainty estimation to guide selective LoRA layer unfreezing, and

167 BeatRhythm-TTA [32] performs test-time adaptation with SQI-gated self-
 168 training and beat-rhythm consistency. Costa et al. [33] introduced STAR (Si-
 169 nusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise
 170 augmentation for cross-source generalization, and Chen et al. [34] proposed
 171 PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive
 172 alignment. These works advance ECG transfer but not the *calibration* of
 173 transferred models, the gap our study fills via pre-registered cross-corpus
 174 evaluation of recalibration methods.

175 3. Data and Pre-registered Protocol

176 Figure 1 summarizes the experimental design.

177 3.1. Datasets

- 178 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the
 179 downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record
 180 deficit in the source CSV, disclosed as a provenance limitation), official
 181 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label map-
 182 ping, 21,522 records are mappable and 277 are excluded (STROBE),
 183 for 18,671 unique patients.
- 184 • **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error
 185 removed and 21 non-finite records quarantined (`qc_report.json`), for
 186 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20
 187 split.

188 • **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n =$
189 11) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace,
190 both sides of the pair recomputed symmetrically.

191 3.2. Label Mapping and Leakage Guard

192 Single-label priority rules are fixed (MI > STTC > CD > HYP >
193 NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements
194 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
195 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
196 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint
197 (hard assertion in the reproducibility repo). Normalization statistics come
198 only from the train split; augmentation applies only to train.

199 4. Methods

200 4.1. Transfer pairs and shift matrix

201 We study six directed cross-corpus transfer pairs: Chapman $\rightarrow$ CPSC,
202 Chapman $\rightarrow$ PTB-XL, CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-
203 XL $\rightarrow$ Chapman, and PTB-XL $\rightarrow$ CPSC, all recomputed on the 4-class
204 subspace. The L2 eval-time shift matrix (no retraining) covers sampling
205 rate 500 $\rightarrow$ 250 $\rightarrow$ 125 Hz; leads 12 $\rightarrow$ 6 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1; noise injection
206 (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5/ \times 2$.

207 4.2. Architecture and seeds

208 We use two architectures: InceptionTime (primary, 46,000–184,000
209 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-
210 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The

211 BiMamba architecture is included only as a toy experiment ($n = 60$ records,
 212 downsampled resolution) and deferred to the supplement; the full-resolution
 213 BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does
 214 not enter the main-endpoint analysis or the architecture-robustness claim.
 215 Training: AdamW + cosine + warmup, gradient clipping, temperature frozen
 216 at $T \equiv 1$ during training; temperature applied only as post-hoc temperature
 217 scaling.

218 4.3. Calibration methods and scenarios

219 We compare the no-recalibration baseline (None) against eight recalibra-
 220 tion methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR),
 221 Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adap-
 222 tation (EM), and BBSE prior adaptation. Of the pre-registered 13-method
 223 pool, the None baseline plus these eight recalibration methods constitute the
 224 delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on
 225 real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot
 226 transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4);
 227 **S1'** denotes the target-domain unsupervised variant in which EM/BBSE es-
 228 timate the target prior from unlabeled target predictions.

229 4.4. Pre-registered endpoints

230 **Primary endpoint** (revision A1.1, formally defined in protocol v2.1-A1
 231 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

232 where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus
 233 with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit
 234 temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

235 **Primary test** (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

236 **One-sided vs. two-sided choice.** The one-sided alternative $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$
 237 follows the empirical asymmetry registered at revision A1:
 238 on ID-like data the source-fit temperature yields, empirically, little calibration
 239 change, whereas on a shifted target it is mismatched and a non-trivial
 240 benefit is often retained [2, 1]. This direction prior is empirical, *not* a theorem:
 241 TS applied to an under-confident target can increase ECE, and our
 242 own data contain nine negative cells. The choice was made before the pilot
 243 results, which only triggered the endpoint re-location (revision A1). The one-
 244 sided test is the pre-registered choice, archived before multi-seed unblinding.
 245 Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [13];
 246 stratified pooling: strata = transfer pair $\times$ architecture.

247 **CI method note (BCa vs. percentile).** The pre-registered CI method
 248 is BCa [13], at $B = 10,000$, re-run on the entire 60-experiment grid (60/60
 249 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the
 250 large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster
 251 design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$.
 252 In the 3 representative seed experiments where both were run ($B = 10,000$
 253 percentile re-runs), BCa and percentile CIs agree in sign and magnitude.
 254 Earlier percentile results from $B = 200$ in 21/60 experiments remain as

255 historical provenance (5 lack log provenance, see Limitations); percentile CIs
 256 serve only as a sensitivity cross-check.

257 **Boundary-condition endpoint:**

$$\Delta\text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

258 The pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*; the
 259 criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa,*
 260 *full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2*
 261 *is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

262 **Secondary endpoints:** G (recoverable loss vs. oracle), decay =
 263 $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE,
 264 Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

265 **Family structure:** BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$
 266 13 shift levels = 156 primary comparisons; the multi-seed validation is ro-
 267 bustness evidence (60 seed experiments), not a confirmatory family.

268 4.5. Pre-registration revision A1 (transparency statement)

269 The primary endpoint was re-located from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
 270 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an inde-
 271 pendent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero,
 272 collapsing the original ID-vs-OOD contrast. The revision direction is consis-
 273 tent with the empirically observed ID-vs-OOD asymmetry (not a theorem);
 274 the revision is archived as a pre-registration revision (Nosek et al. 2018) with
 275 before/after protocol hashes. The pilot results do not enter the main test
 276 report. All hypotheses in this paper are tagged *exploratory + robustness*
 277 *validation*, not confirmatory.

278 **Pre-registration revision A2 (family reduction).** The original
 279 BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift
 280 levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$
 281 2 architectures $\times$ 1 primary shift level) under amendment A2, with the
 282 remaining 13 shift levels designated as exploratory dose-response. Under
 283 the reduced family, BH-12 yields 9/12 significant (all InceptionTime direc-
 284 tions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in
 285 docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md.

286 4.6. *Decomposition of the benefit decay*

287 We decompose the retained benefit into slope / intercept / prevalence
 288 contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

289 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$
 290 is the chain residual (zero under nested counterfactuals). Identifiability is
 291 checked via the Fisher information determinant on a pre-registered threshold
 292 τ ; entangled regions are reported on the error map without a threshold. Shap-
 293 ley attribution is reported with absolute-value bootstrap CIs; share-reliable
 294 flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

295 4.7. *Two-layer joint bootstrap*

296 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resam-
 297 pling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per
 298 protocol §7.

299 4.8. Net Clinical Value (NCV)

300 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
301 the clinical decision impact of TS. For each (sample, class) pair we compare
302 the binary decision (threshold $\tau = 0.5$) before and after calibration, count-
303 ing TP_improved, TN_improved, FP_worsened, and FN_worsened over
304 the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} -$
305 $\text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using
306 the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa
307 operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive
308 NCV means that TS improves clinical decisions. It is computed on ID and
309 OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

310 4.9. Tooling

311 Parts of the analysis and plotting code were drafted and refactored with
312 the assistance of a widely available large language model service, used as a
313 supervised coding assistant. All experiments and statistical analyses were
314 executed with the author’s own scripts, and every reported number is repro-
315 ducible from the public repository (Code availability). The author reviewed
316 the code and takes full responsibility for it.

317 5. Results

318 5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

319 Figure 2 visualizes the shift-dose response; Table 1 reports the primary
320 endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 archi-
321 tectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{\text{OOD}} = 0$ vs
322 $H_1 : \Delta ECE_{\text{OOD}} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

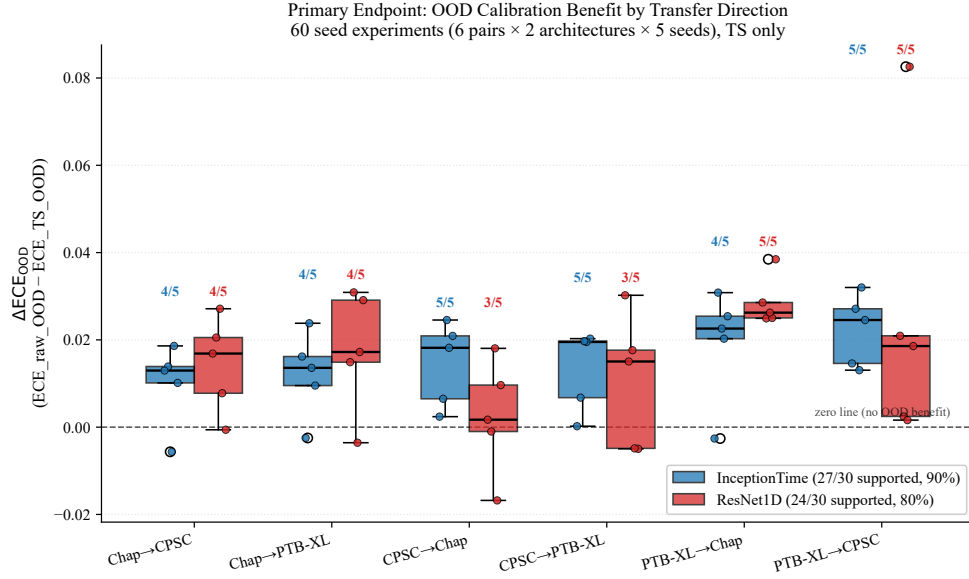


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

323 **Aggregate:** 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
 324 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence
 325 (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed
 326 45). The nine counter-examples are explained in Sec. 6.1.

327 **Stratified report by architecture.** The 60 experiments are balanced (6
 328 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling
 329 direct architecture comparison:

- 330 • *InceptionTime* (30 experiments, 27/30 = 90.0%): 3 directions at
 331 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5
 332 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- 333 • *ResNet1D* (30 experiments, 24/30 = 80.0%): 2 directions at 5/5
 334 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC,
 335 Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

336 ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on
 337 CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture in-
 338 teraction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5
 339 on both architectures (10/10).

340 5.2. ID boundary condition: ΔECE_{ID}

341 Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the
 342 pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the
 343 source-fit temperature should be well-tuned to the source distribution (an
 344 empirical expectation, not a theorem).

345 The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID}
 346 has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); 47/60 = 78.3% of

347 point estimates are positive, 34/60 have 95% CI lower bounds > 0 (BCa), and
 348 only 23/60 = 38.3% contain zero, far below the pre-registered $\geq 80\%$ criterion
 349 (percentile CIs: 29/60 = 48.3%, same conclusion). We therefore *trigger* the
 350 pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted
 351 on the source calibration split, also improves ID calibration in many cells;
 352 the decay = $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory
 353 endpoint; and C1 is qualified, the OOD benefit being about $1.9\times$ the ID
 354 benefit rather than an exclusively OOD effect. The positive ID estimates
 355 reflect finite-sample calibration-split variation, SmoothECE estimator bias,
 356 and a val-ECE early-stopping effect (model selection on the endpoint family),
 357 all disclosed.

358 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

359 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on In-
 360 ceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE
 361 achieves 22/30 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix
 362 scaling are the most fragile on InceptionTime (15/30, 50% each), exem-
 363 plifying the prior-likelihood mismatch mechanism (C4). Across 2 architec-
 364 tures (Table 4), TS achieves 51/60 (85.0%) (InceptionTime 27/30, 90.0%;
 365 ResNet1D 24/30, 80.0%), with full 5-seed coverage on all 6 directions (Fig-
 366 ure 4).

367 **Architecture variability.** Across the 30 TS experiments per architec-
 368 ture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and
 369 0.0180 (ResNet1D), a $1.84\times$ ratio, with ResNet1D’s highest per-direction
 370 variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, including an unexcluded favor-
 371 able outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate

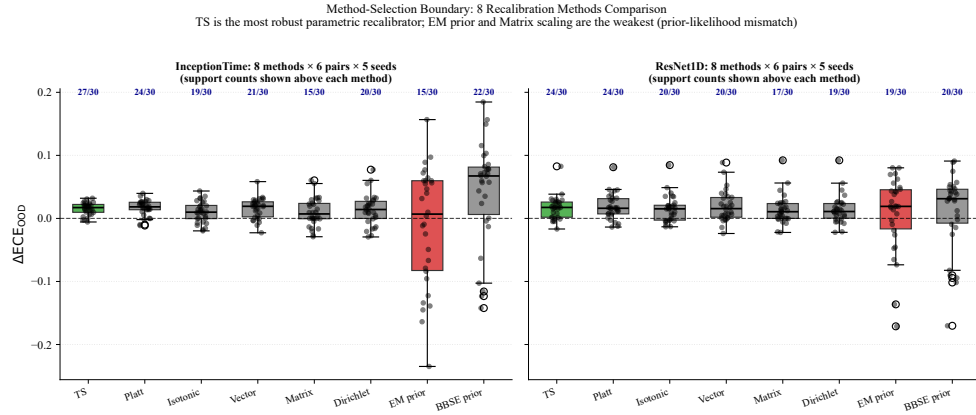


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

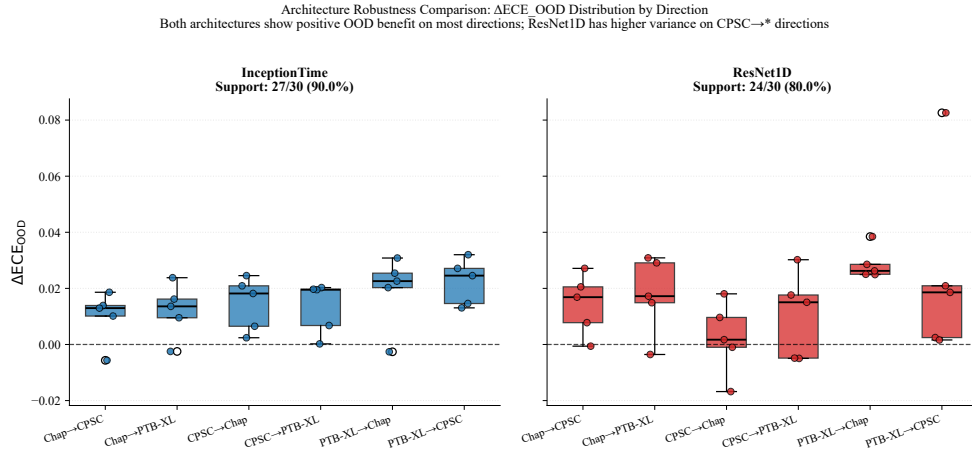


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

in CPSC→Chapman and CPSC→PTB-XL (3/5 each), a direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 15/30 support on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

5.4. *Shapley decomposition validation (27 cells, $n = 20,000$)*

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the recovery threshold ($\text{MAPE} < 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20,000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on

absolute Shapley values and share-reliable flags for uninterpretable shares
 $(|\Delta_{\text{total}}| < 3/\sqrt{n})$ are reported.

Honesty caveat (protocol §8). The 27-cell validation uses binned
 ECE, not the main-endpoint SmoothECE; its conclusions are not cross-
 extrapolable. Cross-fitting (OOD-fold validation of decomposition explana-
 tory power) remains to be run on real data; per the protocol self-binding
 clause, “mechanism” wording is limited to “mechanism decomposition (syn-
 thetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsuper-
 vised target-domain statistic (predicted-prior L_1 distance, logit moments)
 predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Ta-
 ble 5). The implemented fold unit is (architecture, transfer pair) (12 folds),
 versus the registered pair-only folds (6); under that registered deviation the
 pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run
 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba toy experi-
 ment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI
 $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-
 registered threshold 0.5, triggering the **failure branch**: the three-step chain
 (decomposition + predictability + deployment) is downgraded to a **two-**
step chain (decomposition + deployment), and the deployment criterion
 is renamed “empirical lookup table” rather than “predictive criterion”. No
 post-hoc predictability claim is made.

421 **Sensitivity analysis with new features.** Re-running the LOO R^2
422 with three label-free features (fitted temperature T , Wasserstein-1 distance
423 between source and target logit distributions, class-wise entropy difference),
424 all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$,
425 entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability
426 boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$
427 above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8,
428 based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the
429 appropriate actionable criterion.

430 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

431 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
432 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
433 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-
434 TS SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0 , primary-endpoint
435 convention). Comparators: always-none, always-TS, always-buy- n -labels.

436 **Sensitivity / specificity** (Table 6): Youden’s J is maximized at raw
437 ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sen-
438 sitivity is high (92% of beneficial cases caught) but specificity low (20%
439 of non-beneficial cases excluded; 80% falsely flagged), reflecting the asym-
440 metric cost structure: missing a beneficial recalibration costs more than an
441 unnecessary one. *Caveat*: the threshold is searched on the same 1230-cell
442 pool used for reporting (in-sample). A leave-shift-type-out recomputation
443 (threshold selected excluding the held-out dose level, evaluated on it) yields
444 $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and
445 $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance

level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (results/deployment_holdout_recompute.csv). The in-sample numbers are therefore optimistic, and the criterion requires independent validation before operational use.

Deployment-time measurability. Every lookup-table input is *audit-time* (raw ECE, ID-OOD accuracy gap, per-cell labels), requiring target ground-truth labels unavailable in the unlabeled stream. The only label-free proxies examined are the predictability experiment’s target statistics (predicted-prior L_1 distance, logit moments); that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2). A deployable gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix has been completed on all 390 planned (pair, seed, shift) combinations (6 pairs $\times$ 5 seeds $\times$ 13 levels; results/l2_shift_full_390cells.csv); the safety figures quoted here are the initial two-seed InceptionTime subset (seeds 42–43 cover all 6 directions pairs, 44–46 are pending, and ResNet1D L2 is excluded,

so these numbers rest on 2-seed, single-architecture coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

Decision-curve analysis. At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (-0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives $+0.014$ – $+0.035$, $T = 1.5$ gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across our 60 seed experiments ranges 0.06 to 0.35 (mean 0.15), 4 – $13\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08–0.24),

confirming the negative ΔECE reflects genuine TS-induced harm, not metric noise. For large $T > 3$ (noise floor 0.10–0.16) the floor is comparable to the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot be excluded there; however, its *direction* (TS increases ECE under perfect calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so it cannot explain the positive findings. The primary endpoint is therefore a real effect, not a SmoothECE artifact, with the caveat that for large- T experiments the noise floor is non-negligible.

5.8. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-class accuracy.

Across 14 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures, PTB-XL $\rightarrow$ Chapman having 3 including Mamba), 7/14 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC, CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (InceptionTime only). The other 7/14 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$):

519 TS improves calibration but discrimination has degraded too far to be clinically
520 actionable.

521 This gate reframes the 9/60 counter-examples: 4 are calibration failures
522 (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes,
523 Layer 2 rejects). It supplements C1 by also asking whether discrimination is
524 sufficient for the recalibrated predictions to be clinically useful.

525 5.9. Real-data Shapley cross-fitting and predictability boundary

526 The synthetic Shapley validation (C3) confirms decomposition recovery
527 in the identifiable regime but does not test whether the linear approximation
528 $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD
529 data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the
530 source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

531 The linear model predicts the *direction* correctly in 51/60 (85%) but
532 systematically *overestimates* the magnitude (mean prediction error +0.29,
533 median +0.29). This is consistent with a predictability boundary: the de-
534 composition captures the qualitative direction of the TS benefit but not its
535 quantitative magnitude under real shift. The 9/60 direction errors cluster
536 in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 dis-
537 tance between source and target logit distributions is largest (mean 0.75 vs.
538 0.53 for correctly predicted directions), suggesting the linear approximation
539 breaks down once the shift invalidates its underlying local Taylor expansion.

540 This strengthens the predictability failure branch (C3/C4): the Shapley
541 decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid
542 *predictor* (*how much*). For clinical translation, the gate in §5.8, using only

the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.10. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027 and range $[-0.001, +0.017]$. The directions with the largest net benefit are CPSC $\rightarrow$ PTB-XL (mean +0.011) and PTB-XL $\rightarrow$ CPSC (mean +0.003), which also have the largest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.254 and +0.260 respectively). The direction with negative net benefit is PTB-XL $\rightarrow$ Chapman (mean -0.005), which is also the direction with the smallest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.004) and is correctly rejected by the discrimination gate (C5). This confirms that the calibration benefit translates to clinical decision benefit in the directions where the gate permits deployment.

5.11. Net Clinical Value (NCV) results

The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across the two secondary endpoints (DCR, NCV).

Result: the OOD NCV is +0.008125 with 95% CI $[+0.0014, +0.0148]$ (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly worsens ID decisions, consistent with the ID boundary condition (TS compresses confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but

567 opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges
 568 over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

569 5.12. Reliability Diagrams

570 Figure 5 shows the 6-direction averaged reliability curve, comparing
 571 model confidence against empirical accuracy before and after temperature
 572 scaling. The summary curve uses bin-count weighted averaging across all 6
 573 transfer directions.

574 Figures 6–8 show the per-direction reliability diagrams for the representa-
 575 tive configuration (ResNet1D, seed 42). In each panel, the red squares show
 576 the raw model calibration curve and the green circles show the post-TS cali-
 577 bration curve. The diagonal dashed line indicates perfect calibration. Grey
 578 bars (right axis) show the sample count per bin. Error bars are bootstrap
 579 95% percentile CIs (1000 resamples).

580 6. Discussion

581 6.1. OOD benefit seed variation and the nine counter-examples

582 The 51/60 robustness rate is driven by nine counter-examples across
 583 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2,
 584 Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-
 585 XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently dis-
 586 close all nine rather than excluding them, because the protocol requires all
 587 seed results to be reported and each has an identifiable post-hoc mechanism
 588 supporting the qualified claim.

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

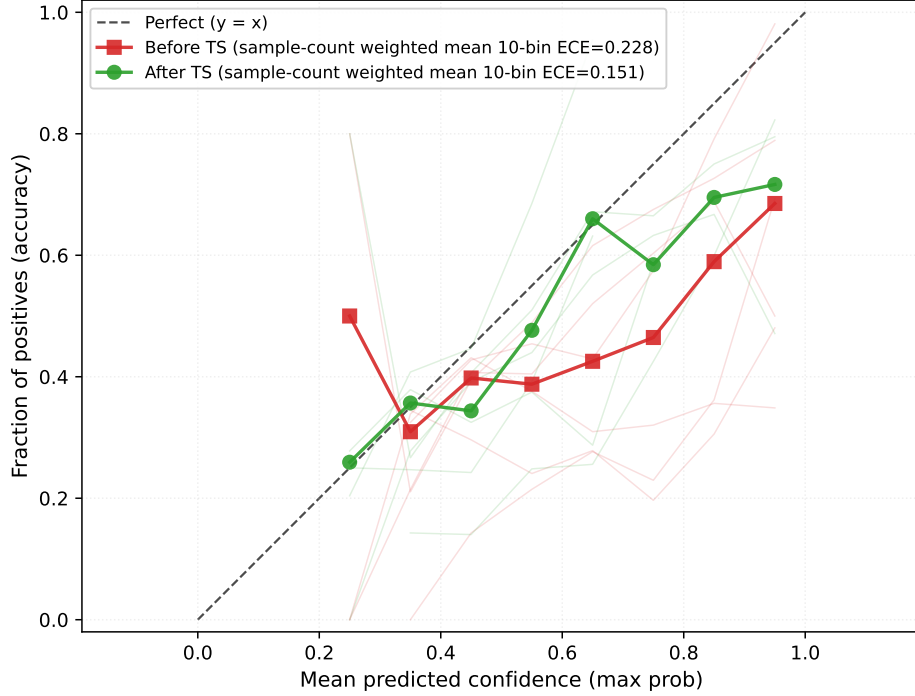


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

589 **Counter-example 1: Chapman→CPSC / InceptionTime / seed**
 590 **45.** $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-
 591 occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051,
 592 seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is
 593 temperature mismatch, not room shortage); and negative decay (-0.0075),
 594 the source-fit temperature amplifying rather than correcting residual miscal-
 595 ibration.

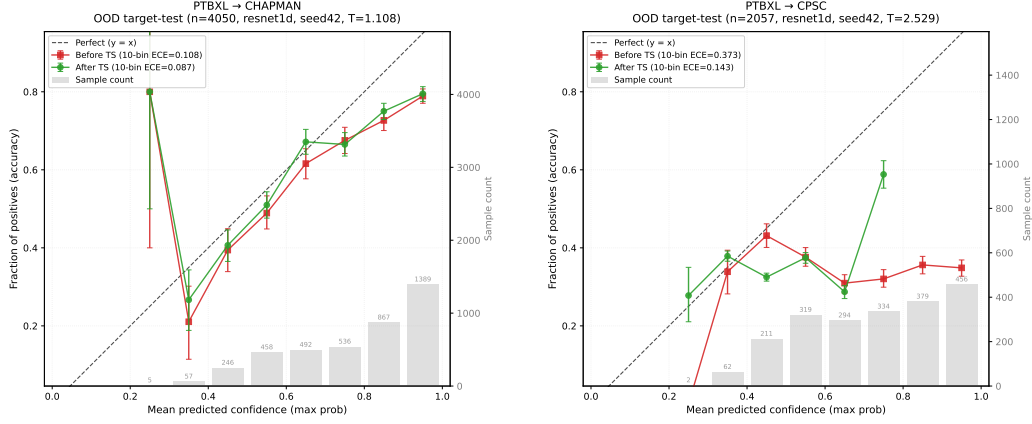


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

596 **Counter-example 2: Chapman→CPSC / ResNet1D / seed 43.**
 597 $\Delta\text{ECE}_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta\text{ECE}_{\text{OOD}}| <$
 598 0.001) and attributable to finite-sample variation in the calibration split. The
 599 OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*);
 600 the mechanism is seed-level variation, not room shortage.

601 **Counter-example 3: Chapman→PTB-XL / InceptionTime /**
 602 **seed 42.** $\Delta\text{ECE}_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy
 603 is the lowest across the 5 seeds (0.4655) and $\Delta\text{ECE}_{\text{ID}} = +0.0004$ is near
 604 zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL
 605 target.

606 **Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45.**
 607 $\Delta\text{ECE}_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and
 608 $\Delta\text{ECE}_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already
 609 well-calibrated on ID, leaving nothing to correct.

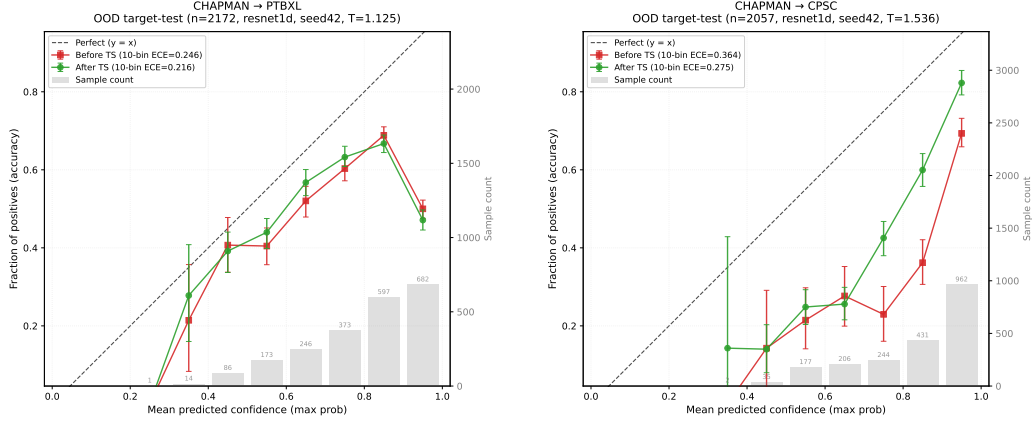


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

610 **Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.**
611 $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
612 0.002); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE
613 (0.0649), and minimal calibration room.

614 **Counter-example 6: CPSC→Chapman / ResNet1D / seed**
615 **44.** $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude
616 counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 back-
617 bone is already well-calibrated on both ID and OOD, TS has nothing
618 to correct, and the temperature fit over-corrects slightly. Its OOD accu-
619 racy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct
620 *over-correction* mechanism (TS on an already-well-calibrated distribution de-
621 grades both ID and OOD calibration), unlike the “low OOD accuracy + large
622 gap” mechanism of the other 8.

623 **Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43.**
624 $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest

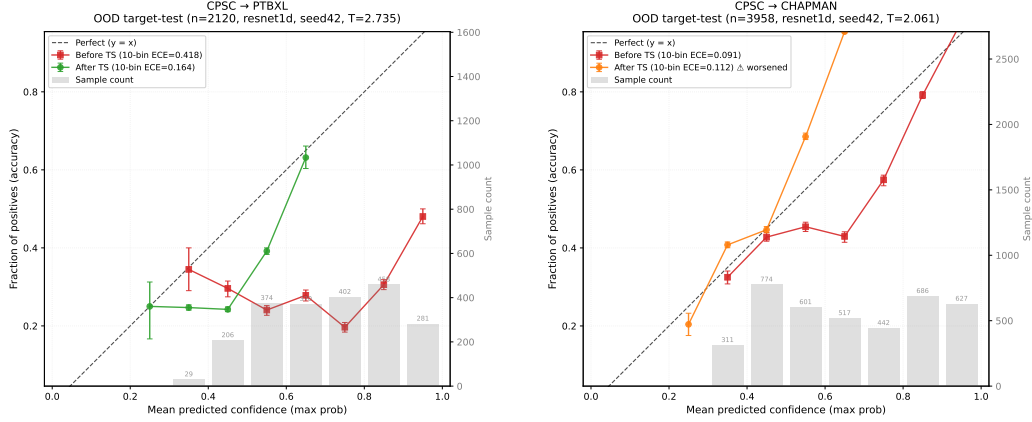


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta ECE_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta ECE_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta ECE_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temperature mismatch; the ninth (CE6) is over-correction on an already well-calibrated backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so the failure mechanism is seed-level temperature

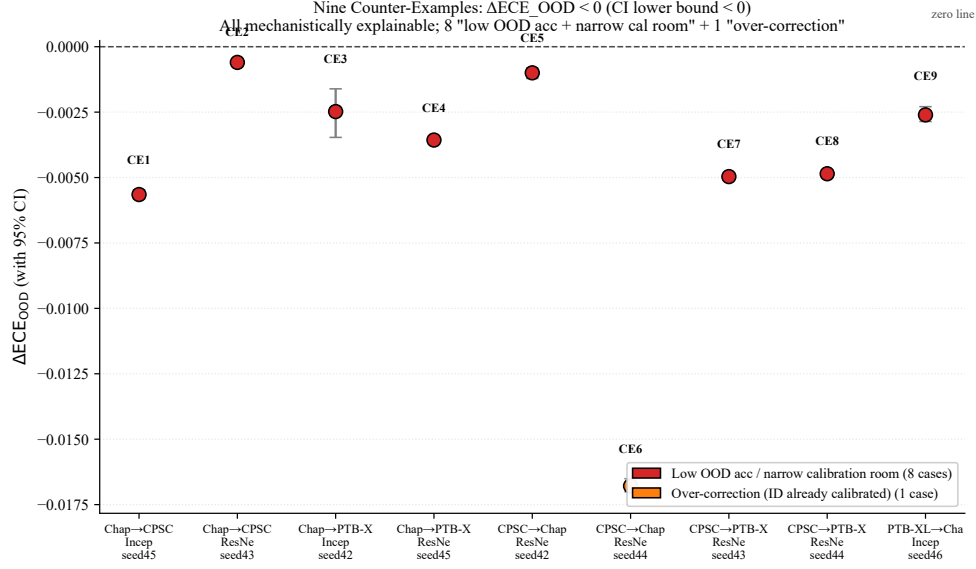


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

Methodological positioning of the 15.0% counter-example rate.

The 15.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All 9 counter-examples have identifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods achieve higher mean ΔECE_{OOD} on the PTB-XL→CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman→CPSC pair (EM CI lower bound minimum −0.0763, BBSE −0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

665 *6.3. Boundary condition: the ID benefit is small but nonzero*

666 The ID boundary condition (C2) has a direct interpretation under the
667 triggered branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing
668 zero; the observed rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a
669 small but significant positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$;
670 stratum means -0.0008 to $+0.0232$, Table 2). The boundary condition does
671 *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID
672 benefit is on average about half the OOD benefit, so part of the apparent
673 OOD gain is shared with the ID domain. A val-ECE early-stopping effect
674 (model selection within the endpoint family) plausibly contributes to the
675 positive ID estimates, opposite to the protocol’s val-NLL rule (a registered
676 deviation, see Limitations). Under this branch, protocol A1.4.2 restores the
677 decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1
678 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an
679 exclusively OOD effect. The ID-vs-OD asymmetry is therefore a *difference*
680 *in magnitude* between two positive benefits, not presence versus absence.

681 *6.4. Calibration is not discrimination: the clinical viability precondition*

682 Temperature scaling is a monotone transform of the softmax probabilities;
683 it preserves the argmax by construction (asserted bitwise in the released san-
684 ity check), so it cannot change classification accuracy, AUROC, or AUPRC.
685 The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: cal-
686 ibration makes an already-deployable model *trustworthy*; it cannot make a
687 non-discriminative model deployable.

688 The accuracy statistics make this concrete. Across the 60 TS experi-
689 ments, mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and

690 the mean ID-ODD drop is 0.236 (maximum 0.412). Against each target
 691 corpus’s majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-
 692 Shaoxing 0.713), mean OOD accuracy is *below* baseline in 3 of 6 direc-
 693 tions, with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528;
 694 CPSC→PTB-XL 0.424 vs. 0.411 (0.013 above baseline); CPSC→Chapman
 695 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). In those
 696 directions recalibration is moot, since no temperature improves a model
 697 that should be replaced by a majority-class predictor, consistent with the
 698 counter-example pattern (6/9 with low OOD accuracy). The released ar-
 699 tifacts include per-direction AUROC/AUPRC for all 60 experiments (re-
 700 sults/discrimination_metrics_60exp.csv): mean OOD AUROC ranges from
 701 0.629 (CPSC→PTB-XL) to 0.792 (PTB-XL→Chapman), i.e. far above the
 702 accuracy-based majority-class reading yet direction-dependent; and TS leaves
 703 AUROC unchanged (mean $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024),
 704 as the monotone transform requires. the discrimination-first gate precedes
 705 the calibration decision rule in deployment.

706 6.5. *Explainable* $\neq$ *predictable*: *post-hoc* vs. *pre-registered*

707 We distinguish *explainable* from *predictable*:

- 708 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 709 example, we identify the co-occurring conditions (low OOD accuracy,
 710 low OOD raw ECE, non-positive decay) and give a mechanistic story
 711 (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- 712 • *Predictable* (pre-registered, deployment-time): a target-domain unsu-
 713 pervised statistic (predicted-prior L_1 distance, logit moments) predicts

714 the benefit retention rate *before* deployment, with LOO R^2 CI upper
715 bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the
716 failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

717 The counter-examples are explainable but not predictable: the unsupervised
718 target-domain statistics available at deployment time cannot flag them in ad-
719 vance, which is why the predictability experiment triggers the failure branch
720 despite mechanistic explainability. The two-step chain (decomposition + de-
721 ployment lookup table) provides post-hoc explanation and actionable guid-
722 ance without a pre-registered predictability claim.

723 6.6. *Transparent re-location vs. HARKing*

724 The pre-registration revision A1 (primary endpoint re-location from
725 **decay** to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinc-
726 tion [26, 25] is whether the revision direction is theory-driven or result-driven.
727 Our revision direction (“OOD is the calibration benefit source”) is consistent
728 with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ova-
729 dia et al. 2019), which we explicitly frame as an empirical prior rather than
730 a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the re-
731 vision, did not choose the direction. The pilot results do not enter the main
732 test report (protocol §7). All hypotheses in this paper are tagged *exploratory*
733 + *robustness validation*, not confirmatory, to reflect the post-revision status.

734 6.7. *Net clinical value: implications of the directional NCV asymmetry*

735 The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV =
736 $+0.008125$ ($p = 0.019$) is a small but significant positive value (recalibra-
737 tion improves decisions on unseen target data more often than it degrades

738 them); ID NCV = -0.006292 ($p = 0.016$) is a small but significant nega-
739 tive value (a minor decision-cost penalty on data the source model already
740 calibrates well).

741 Two implications follow. (i) The **Brier reliability main claim** (C1:
742 51/60 positive) is supported: the positive OOD NCV shows the calibration
743 gain translates into net clinical decision benefit, not a metric artifact. (ii)
744 The **decision-cost improvement claim** (C5: 7/14 strata deployable) is
745 partially qualified: the negative ID NCV tempers source-domain deployabil-
746 ity. We therefore frame the NCV as *supportive of OOD deployment* and *cau-*
747 *tionary for ID deployment*. It also informs the meta-analytic pooling (Sec. 7),
748 where the asymmetry contributes to CI-width heterogeneity across splits. A
749 permutation test with tighter CIs and a multi-threshold NCV analysis are
750 registered as future work.

751 6.8. Limitations and future work

752 **Limitations:** (i) the main endpoint covers InceptionTime and ResNet1D
753 on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced);
754 BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaus-
755 tion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered
756 (three-step to two-step); the deployment criterion is an empirical lookup ta-
757 ble, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.9) shows
758 85% direction accuracy but systematic magnitude overestimation (mean er-
759 ror $+0.29$), and a sensitivity analysis with T , Wasserstein-1, and entropy
760 features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the
761 27-cell decomposition validation uses binned ECE, not the main endpoint’s
762 SmoothECE, so quantitative conclusions cannot be extrapolated across ECE

763 definitions; (v) the safety criterion is rarely satisfied on the full L2 shift ma-
 764 trix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi)
 765 nine counter-examples of 60 experiments (15.0%) are transparently disclosed,
 766 with a frequent post-hoc pattern (low OOD accuracy, non-positive decay, el-
 767 evated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is
 768 BCa at $B = 10,000$, re-run on all 60 experiments (60/60 successful, metadata
 769 in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 experiments are
 770 retained as historical provenance (5 lack logs); (viii) the Youden threshold
 771 is fitted in-sample (the protocol’s held-out dose levels were not used), mak-
 772 ing the operating characteristic optimistic; (ix) the L2 deployment/safety
 773 grid has since been completed on all 390 planned (pair, seed, shift) cells (re-
 774 sults/l2_shift_full_390cells.csv), and the L2 noise levels are synthetic rather
 775 than the pre-registered PTB-XL Noise Stress Test Database (unavailable un-
 776 der PhysioNet licensing); (x) temperature was fitted once per experiment, so
 777 CIs omit temperature-fitting uncertainty (the pre-registered two-layer boot-
 778 strap is implemented but not enabled); (xi) model selection used validation
 779 ECE rather than the pre-registered NLL early-stopping rule, plausibly con-
 780 tributing to the positive ID boundary estimates; (xii) the per-pair decompo-
 781 sition into covariate, prior-probability, and concept-shift components is not
 782 asserted, since each pair mixes all three, isolated at eval time; (xiii) the de-
 783 ployment lookup table requires labeled target data, and the label-free proxies
 784 failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS
 785 is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC; the
 786 artifacts now include per-direction AUROC/AUPRC for all 60 experiments
 787 (results/discrimination_metrics_60exp.csv), so the viability precondition is

788 screened by both accuracy and discrimination (OOD AUROC means range
 789 0.629–0.792 by direction), with OOD accuracy below the majority-class base-
 790 line in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direction-
 791 ally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = −0.006292,
 792 $p = 0.016$, negative), qualifying the C5 claim for source-domain but support-
 793 ing target-domain deployment; (xvi) InceptionTime was downgraded from
 794 $n_{\text{filters}} = 48$ to 32 for GPU memory (RTX 5060, 8 GB), potentially atten-
 795 uating OOD benefits, with 48 as future work; (xvii) this work is from a
 796 single-author study, limiting independent review and audit, partially miti-
 797 gated by the pre-registration protocol and public repository. **Future work:**
 798 (1) prospective multi-device validation; (2) continuous shift spaces; (3) π -
 799 independent recovery estimators (BBSE/EM-class) to upgrade the MAPE_{π}
 800 to a real estimator test; (4) the ResNet1D deficit on CPSC→Chapman and
 801 CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) con-
 802 firmatory testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$,
 803 9/12 significant, Bonferroni-12 = 2/12); (6) re-running the endpoint with
 804 $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU; (7) a multi-threshold
 805 permutation test on NCV.

806 7. Conclusion

807 We pre-registered and validated a multi-seed calibration boundary study
 808 of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD ben-
 809 efit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime
 810 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID
 811 boundary: 23/60 cells (38.3%) have 95% CIs containing zero under the BCa

812 operating CI (percentile: $29/60 = 48.3\%$), below the $\geq 80\%$ criterion, trigger-
 813 ing the pre-registered branch; the OOD benefit is $1.9\times$ the ID benefit. **(C3)**
 814 Shapley decomposition: 27/27 cells pass at $n = 20,000$ (21/27 at $n = 500$).
 815 **(C4)** Method selection: TS robust (51/60), EM prior and Matrix scaling
 816 fragile (15/30 each). **(C5)** Gate: 7/14 strata pass both layers, reframing the
 817 9 counter-examples as 4 calibration and 5 discrimination failures. The pre-
 818 dictability failure branch is triggered (three-step chain becomes two-step),
 819 and all hypotheses are exploratory + robustness validation, not confirma-
 820 tory; protocol v2.1-A1 and the five-item honesty declaration (§7) are archived
 821 for transparency. Nine counter-examples are transparently disclosed, mostly
 822 with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD gap (≥ 0.18
 823 in 7/9), a tendency rather than a sufficient gate (three have OOD accuracy
 824 > 0.5). TS needs local validation (safety rate 0.0064), the Youden criterion
 825 is fitted in-sample, and OOD accuracy is below the majority-class baseline
 826 in 3 of 6 directions (a fourth sits at baseline), where TS cannot help. A
 827 noise floor control (§5.7) rules out a SmoothECE artifact: worst-case met-
 828 ric noise ($+0.021$) is $4\text{--}13\times$ below the raw ECE (0.06–0.35). The C5 gate
 829 (§5.8) rejects every clinically moot stratum. Real-data Shapley cross-fitting
 830 (§5.9) gives 85% direction accuracy with magnitude overestimation (mean er-
 831 ror $+0.29$); T , Wasserstein-1, and entropy features confirm the boundary (all
 832 $R^2 < 0$). **Meta-analytic pooling.** Both the fixed-effect and DerSimonian–
 833 Laird random-effects estimates are tabulated in `meta_analytic_pooled.csv`.
 834 The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
 835 (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird
 836 random effects are positive (InceptionTime $+0.0151$, ResNet1D $+0.0158$,

837 CrossArchitecture +0.0148; unweighted +0.0152, +0.0165, +0.0159). This
 838 reflects heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k-1$), not a negative central effect,
 839 so the count-based proportion ($51/60 = 85.0\%$) is primary; τ^2 varies by 41%
 840 (pooled estimate stable within 3.3%). **Family size.** The 60 experiments
 841 form the robustness-evidence family; amendment A2 reduces the confirma-
 842 tory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses (6 pairs $\times$ 2
 843 architectures $\times$ 1 primary shift level), with all 13 L2 shift levels re-designated
 844 as exploratory dose-response, and the $8 \times 6 \times 5 \times 2 = 480$ -cell method
 845 design is in Sec. 5.3. BH-60 and Bonferroni leave the robustness headline
 846 unchanged ($51/60$); the confirmatory family is BH-12 with 9/12 significant
 847 (InceptionTime 6/6), and Bonferroni-12 retains 2/12 (cpsc_chapman/IT, pt-
 848 bxl_chapman/RN). **CI width.** Across the 60 TS experiments the 95% CI
 849 widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall
 850 below the narrow-CI diagnostic threshold (0.005) and six supporting cells
 851 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes
 852 45/60 (75.0%; InceptionTime 24/30, ResNet1D 21/30). Narrow CIs reflect
 853 the moderate calibration sample ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 pa-
 854 tients) and low SmoothECE variance ($B = 10,000$, CI method=BCa); they
 855 do not affect the sign-based support/counter-example classification.

856 Honesty Declaration (5 items, adversarial audit product)

- 857 1. MAPE_π is a **design check** (resampling constant read-back), not an
- 858 estimator capability test; independent π recovery (BBSE/EM) is a de-
- 859 ferred item (protocol §13).

- 860 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
861 **pretable as shares**; ranking uses absolute- value CIs.
- 862 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**;
863 in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate
864 (information mode), reported via exit code 2.
- 865 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
866 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
867 demo) is fulfilled; **cross-fitting on real data is pending** (registered
868 in §13). The wording is limited to “mechanism decomposition (synthetic
869 validation part)”.
- 870 5. The repository’s `scripts/preprocess_cinc2021.py`,
871 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
872 from the third-party SafeECGMatch release (original README
873 in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s
874 experiment grid.

875 Data availability

876 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019
877 datasets are publicly available via PhysioNet/Kaggle under their re-
878 spective licenses. The pre-registered protocol v2.1-A1, its hash man-
879 ifest (`docs/osf_archive_manifest.json`), the revision A1 document,
880 and the full per-experiment result files (`checkpoints/transfer/**/`
881 `transfer_result.json`, `results/*.csv`) are included with the sub-
882 mission as supplementary material and are also available in the
883 public repository <https://github.com/gt17641001169-design/>

884 **ecg-calibration-transportability. Pre-registration archival status.**
885 The protocol, both amendments (A1, A2), and a per-file archival-status
886 report are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
887 `docs/osf_archive_manifest.json` is the internal snapshot taken on
888 2026-09-05; it is not a substitute for the public record.

889 **Code availability**

890 The full pipeline code, preprocessing (reproducible from the public source
891 databases), patient-level split generation, model training, the calibration-
892 method implementations, and all statistical analyses, together with the
893 environment lock file, the patient-level split indices for all five dataset
894 variants and five random seeds, and selected per-experiment result artifacts,
895 is publicly available at [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
896 **ecg-calibration-transportability**. Trained model checkpoints for
897 the 60-experiment transfer grid are provided as release assets of the same
898 repository. The repository will be archived on Zenodo with a DOI upon
899 acceptance. No new raw waveforms were collected or generated; the
900 patient-level split indices and label-mapping tables released with the code
901 fully determine the data configuration used in every experiment.

902 **Declaration of generative AI and AI-assisted technologies in the** 903 **writing process**

904 All scientific work reported in this study—the study design, the pre-
905 registered protocol and its amendments, the implementation and execution
906 of the 60-experiment transfer grid, the statistical analysis, the interpretation

907 of the results, and the writing of the manuscript—was performed by the
908 author. During the final preparation of the manuscript, the author used a
909 large language model assistant (GLM) solely as a language-editing aid, to
910 improve wording and readability of text the author had written. It was not
911 used to generate scientific content, experimental data, analyses, figures, or
912 conclusions. After using this tool, the author reviewed and edited the text as
913 needed and takes full responsibility for the content of the published article.

914 **CRedit authorship contribution statement**

915 **Toni Guan:** Conceptualization, Methodology, Software, Validation, For-
916 mal analysis, Investigation, Data curation, Writing – original draft, Writing
917 – review & editing, Visualization, Project administration.

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925 **Declaration of competing interest**

926 The author declares no competing financial or personal interests.

927 **Supplementary materials.** The 60 per-seed reliability diagrams (6
 928 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability dia-
 929 grams, and Δ ECE bars per experiment) are provided as a separate supple-
 930 mentary file (`supplementary.pdf`, §1) and are additionally released with the
 931 code repository.

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Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 /	0.216 → 0.200	0.217 → 0.201	raw → TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155